| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2022 | **Task Last Updated:** | FY 09/08/2022 |
| **PI Name:** | Shepherd, John Ph.D. | | |
| **Project Title:** | Space-Feasible Body Composition and Body Shape Analysis for Long Duration Missions | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | TRISH--TRISH | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | johnshep@hawaii.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | |
| **Organization Name:** | University of Hawaii | | |
| **PI Address 1:** | 701 Ilalo Street | | |
| **PI Address 2:** | Suite 522 | | |
| **PI Web Page:** | | | |
| **City:** | Honolulu | **State:** | HI |
| **Zip Code:** | 96813 | **Congressional District:** | 1 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | TRISH--Focused Investigations |
| **Start Date:** | 09/01/2019 | **End Date:** | 08/31/2021 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | TRISH |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 8/31/2021 per TRISH (original end date was 8/31/2020)--Ed., 7/9/20 | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Sadowski, Peter ( University of Hawaii ) | | |
| **Grant/Contract No.:** | NNX16AO69A-FIP0017 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

| | |
|---|---|
| **Task Description:** | Focused Investigation Project<br>We propose to monitor frailty risk using 3D optical scans that have adjustments for fluid redistribution. 3DO models accurately estimate bone and body composition but lack space acclimation experience. We will perform studies to select hardware, algorithms, and augment models with microgravity analogs. We conclude with making a space-feasible prototype for microgravity testing during parabolic flights.<br><br>Aim 1: Determine the optimal performance and space feasibility of 3D optical cameras to collect views of the body by characterizing the resolution, framing rate, precision of acquisition, and contrast detail characteristics using anthropomorphic phantoms.<br><br>Aim 2: Explore and identify the accuracy and precision of 3DO derived total body composition (lean, fat, percent fat, bone mineral density/BMD), special regions (visceral fat, subcutaneous fat, lumbar spine BMD), and automated anthropometry using pose-varied, limited-view scanning, analyzed with pose removed, compared to criterion methods (dual-energy x-ray absorptiometry/DXA and high-resolution 3DO).<br><br>Aim 3: Identify accuracy and precision limitations of 3DO body composition and automated anthropometry due to effects of space acclimation using surrogate conditions (poses, postures, inverted gravity (inversion boots), buoyancy (underwater) and microgravity (trampoline apex)).<br><br>Aim 4: Construct and describe the performance of a space-feasible prototype under microgravity conditions during parabolic flights.<br><br>Aim 1 Deliverables: Measures of performance of multiple camera candidates include the following: single frame spatial resolution at a different fields of view and focal lengths (Nyquist criterion), landmarks detection in terms of contrast detail curves (3) (i.e., grid of progressively larger diameter and higher contrast markers like Avery stickers), repeat of the above measures in progressively lighter rooms (lumens background), above measures in presence of multiple cameras operating at the same time, power consumption, footprint. Cameras will be ranked for each measure.<br><br>Aim 2 Deliverable: Accuracy and precision estimates (compared to criterion measures) of 30 subjects scanned using the candidate cameras to derive body composition and body shape estimates: a) in a variety of poses but algorithmically unposed, b) whole-body unposed scans created from limited field of view scans, c) shape and appearance models of the head/neck. The measurement estimates include the following: total body fat; lean, percent fat; appendicular lean mass; visceral and subcutaneous fat; total BMD; lumbar spine BMD; waist, hip, thigh, arm, neck circumferences; body dimensions of torso, arm, leg; total length; body surface area; body volume; and head/neck shape and appearance using Cootes methods (4-6).<br><br>Aim 3 Deliverables: Accuracy and precision estimates of the 3DO models for scans that include body shape altering effects that mimic space acclimation including (poses, postures, inverted gravity (inversion boots), buoyancy (underwater), and microgravity (trampoline apex) restrictive elastic clothing). Two approaches will be tested: conventional models as tested in Aims 1 and 2, as well as a deep learning model.<br><br>Aim 4 Deliverables: Accuracy and precision of the measures from Aim 2 during weightlessness (freefall). Delivery of a least significant change and monitoring time interval strategy for use of 3DO in space for each of the Aim 2 measures. List of remaining challenges to study before space-based testing.<br><br>Note that we will not be able to test the effects of fluid redistribution within this application. Future studies will include collaborations with acute head-tilt studies to study fluid redistribution effects on 3DO accuracy and our proposed method to mitigate the effect using head/neck imaging. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This project has shown the utility of human body scanning for quantifying body composition for space-based environments. Body composition including fat and muscle masses of the total body, arms, and legs can be quantified during spaceflight using the low-power small-footprint systems. |
| **Task Progress:** | Fiscal Year 2022 Update Research efforts have been captured in the following peer-reviewed publication (Ed., 12/31/21): Wong MC, Ng BK, Tian I, Sobhiyeh S, Pagano I, Dechenaud M, Kennedy SF, Liu YE, Kelly NN, Chow D, Garber AK, Maskarinec G, Pujades S, Black MJ, Curless B, Heymsfield SB, Shepherd JA. "A pose-independent method for accurate and precise body composition from 3D optical scans." Obesity (Silver Spring). 2021 Nov;29(11):1835-47. <a target="_blank" href="https://doi.org/10.1002/oby.23256">https://</a> ; PubMed <a target="_blank" href="http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=pubmed&dopt=Abstract&list_uids=34549543">PMID: 34549543</a>; PubMed Central <a target="_blank" href="http://www.ncbi.nlm.nih.gov/pmc/articles/PMC8570991">PMCID: PMC8570991</a><br><br>###<br><br>Aim 1 Conclusion: After extensive testing, we used the parameters that provided the best 3DO images. The Intel RealSense D435 was the best available camera for our 3DO imaging. The depth field of view for the D435 allows for scanning in small spaces because the sensor can get closer to the subject while retaining their entire body. In the experiments conducted, we found that using three D435 sensors allowed for better coverage of the subject and resulted in a more complete mesh. Aim 2 Conclusion: The three-camera SRL system performed with accuracy and precision in respects to criterion measures, Fit3D and DXA, in standard pose positioning. When pose was altered or when using a partial mesh, accurate body composition results were also successfully derived. Aim 3 Conclusion: The inverted scans had strong agreement with criterion measures for lean mass estimates. Inverted test-retest precision for lean estimates were similar to the standing SRL precision. Aim 4 Conclusion: The one-camera system performed as well as the 3-camera system. The 1-camera system is more accessible and can be easily transported to different locations. However, the space required to capture a scan would be larger than the 3-camera SRL system because the 1-camera system requires users to scan subjects from a certain distance to ensure the subject is in the field of view. |
| **Bibliography Type:** | Description: (Last Updated: 12/27/2024) |

| | |
|---|---|
| **Articles in Peer-reviewed Journals** | Wong MC, Ng BK, Tian I, Sobhiyeh S, Pagano I, Dechenaud M, Kennedy SF, Liu YE, Kelly NN, Chow D, Garber AK, Maskarinec G, Pujades S, Black MJ, Curless B, Heymsfield SB, Shepherd JA. "A pose-independent method for accurate and precise body composition from 3D optical scans." Obesity (Silver Spring). 2021 Nov;29(11):1835-47. https://doi.org/10.1002/oby.23256 ; PubMed PMID: 34549543; PubMed Central PMCID: PMC8570991 , Nov-2021 |
| **Awards** | Shepherd J. "American Institute for Medical and Biological Engineering (AIMBE) College of Fellows Award." Mar-2021 |